

IoT Design (Physical/Logical)

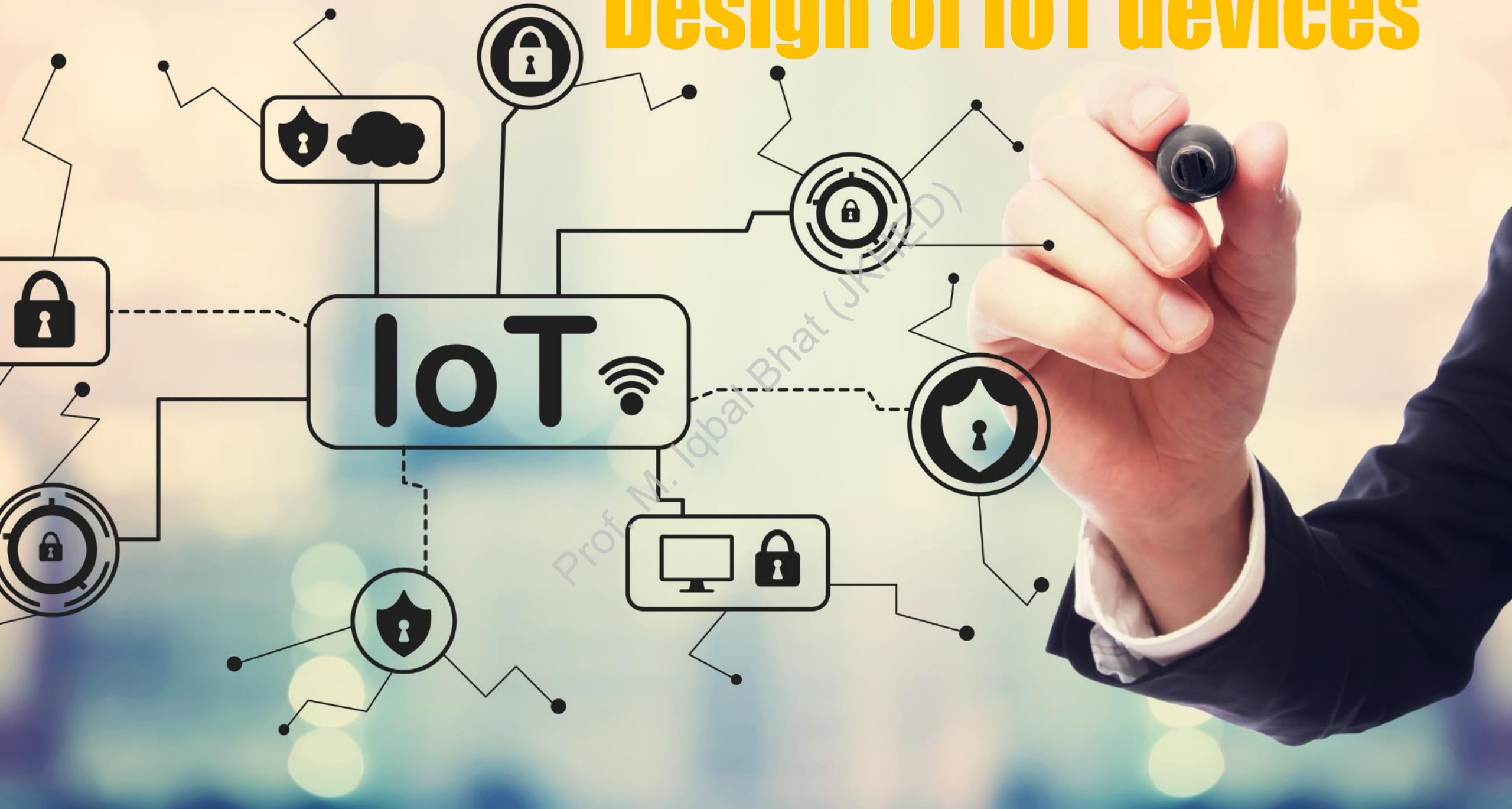


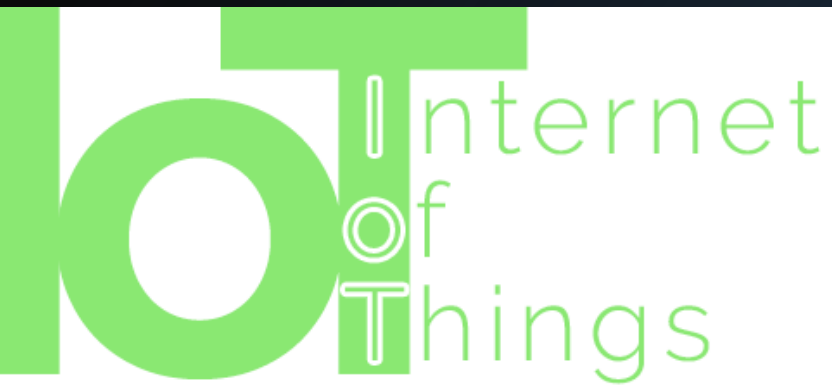
Mohd. Iqbal Bhat

Assistant Professor,

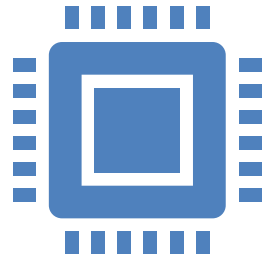
Government Degree College Beerwah

Design of IoT devices





Design of IoT Devices



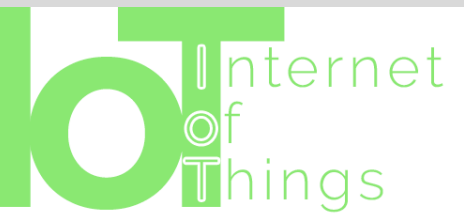
Physical Design:

Involves the actual hardware such as sensors, actuators, node devices and protocols



Logical Design:

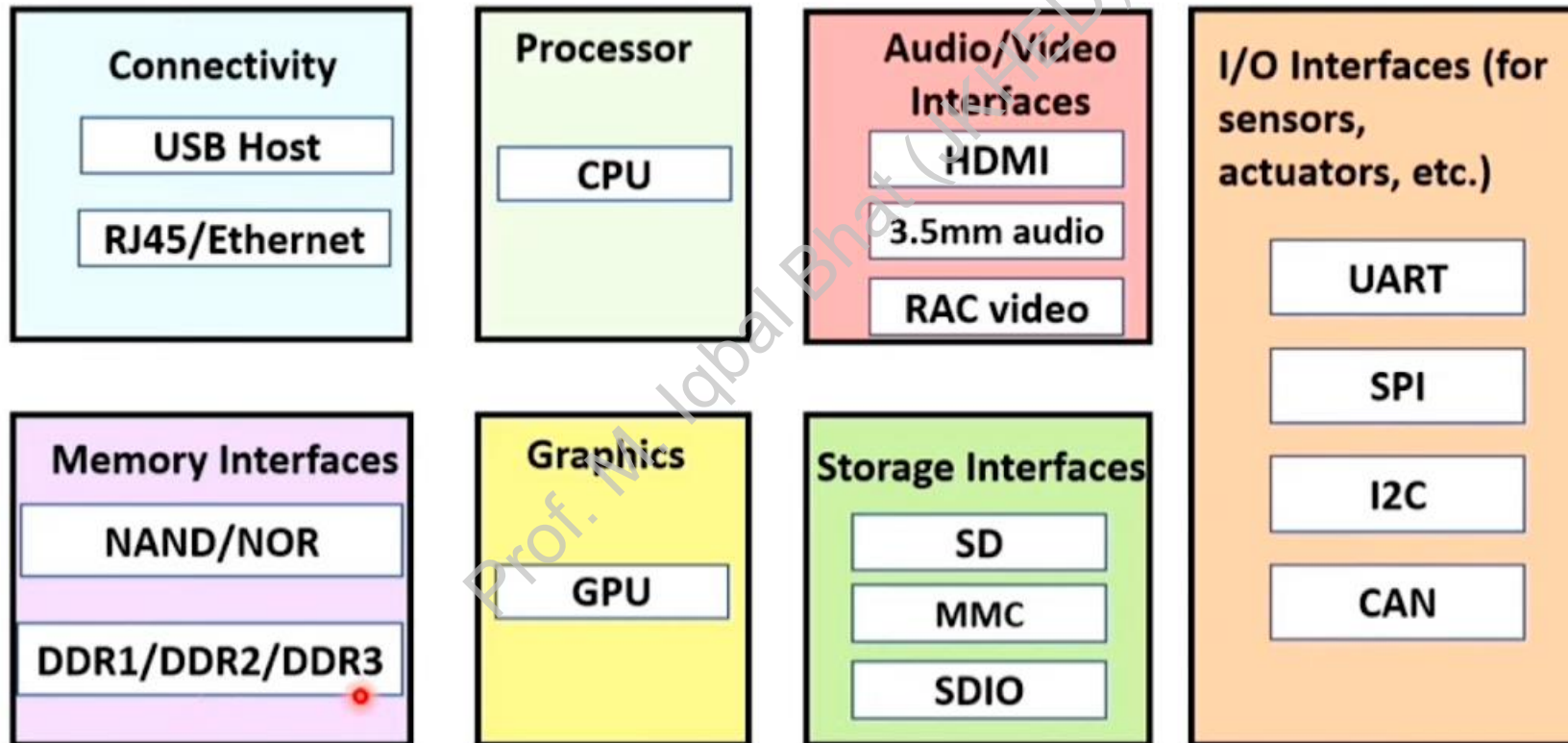
Represents the conceptual model of the system, outlining how devices interact and process data



Physical Design of IoT Devices

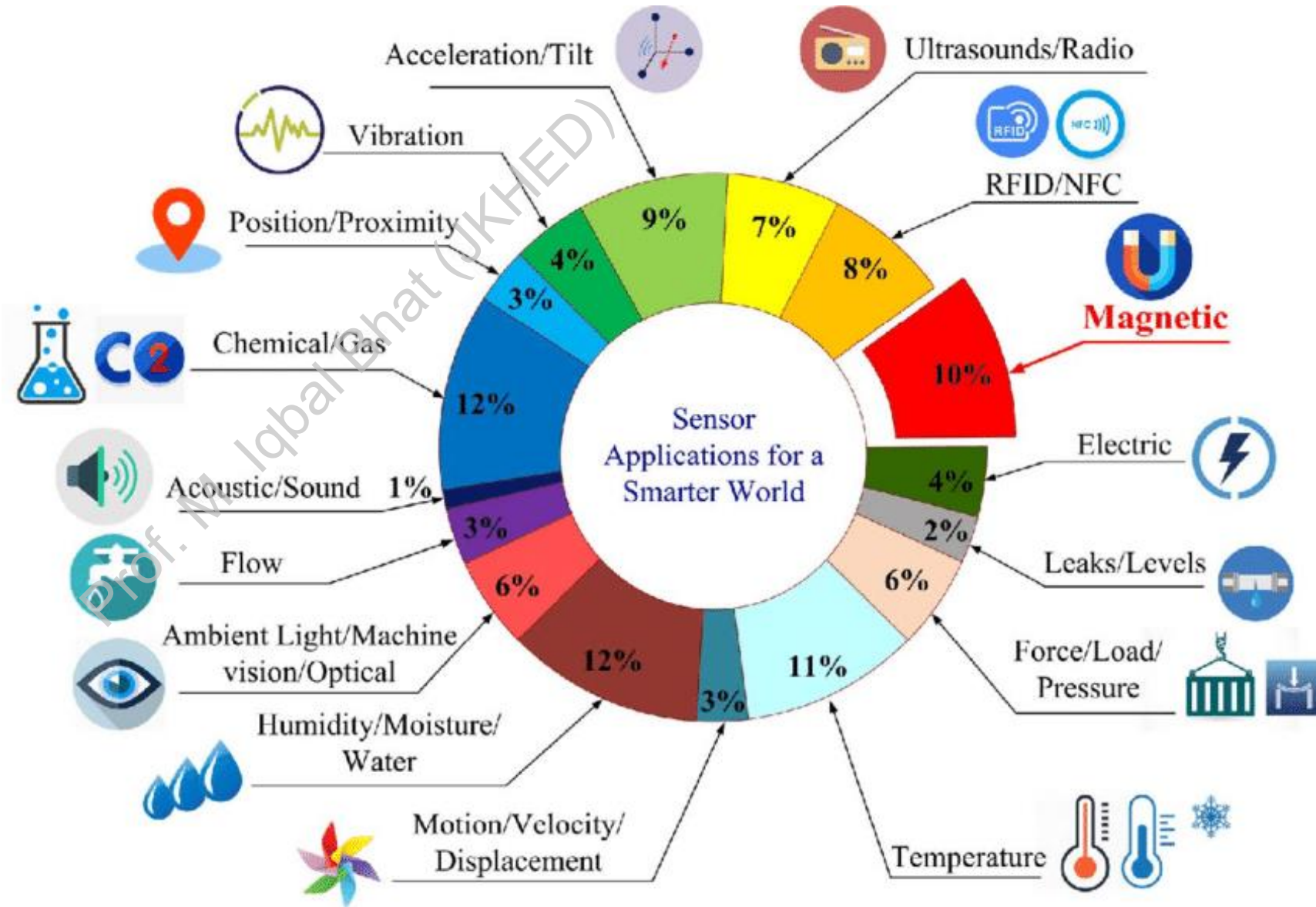
- It includes:
 - **Things/Devices**
 - The things in IoT refers to IoT Devices which have unique identities and can perform remote sensing, actuating and monitoring capabilities.
 - Can collect and exchange data.
 - Can perform other tasks within the IoT infrastructure.
 - **Protocols:**
 - Foundation of IoT Systems
 - Allow devices to exchange data over the network
 - Define data exchange formats, data encoding, addressing schemes for device and routing of packets from source to destination
 - Controls sequence control, flow control and retransmission of packets.

Block diagram of IoT Devices



Building blocks of IoT

- IoT Devices / Nodes
 - Sensors
 - Objects
 - RFID Tags
 - Actuators

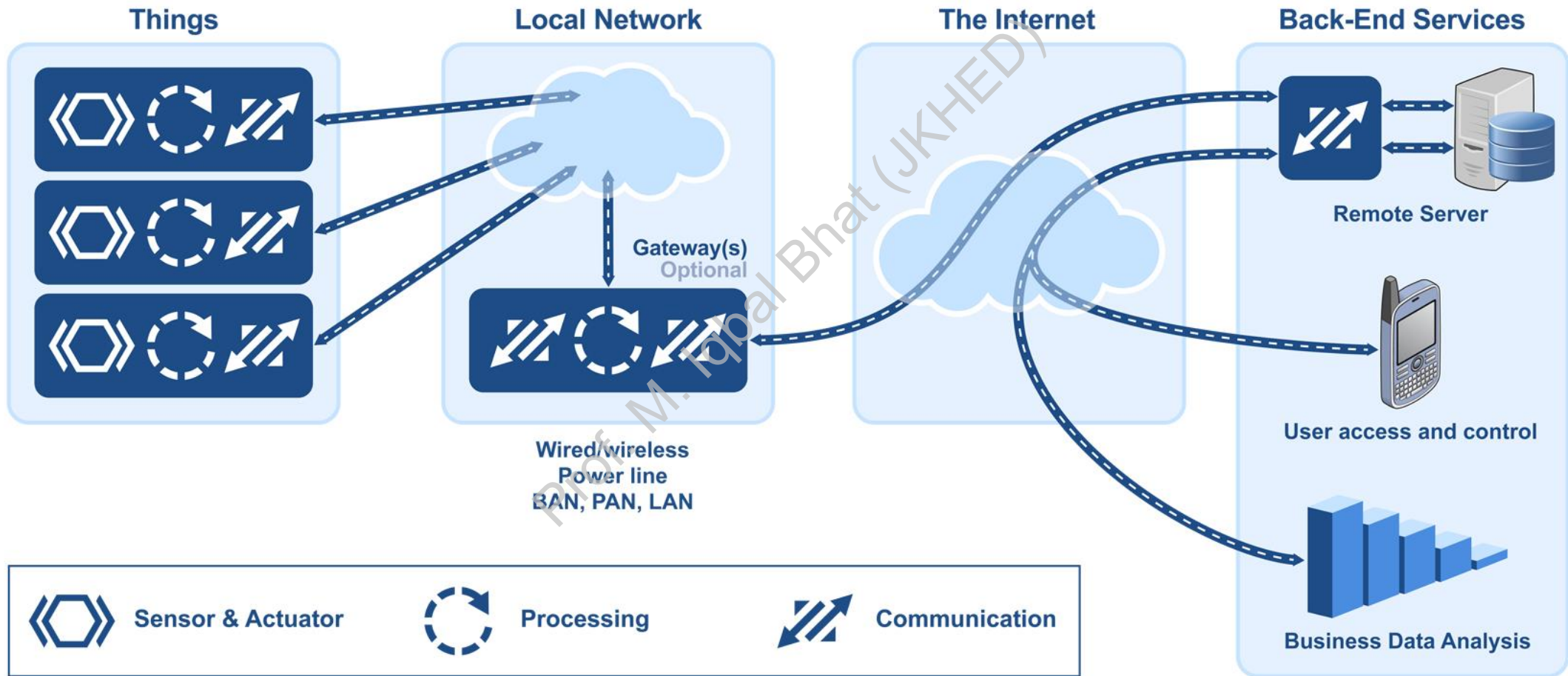


Building blocks of IoT

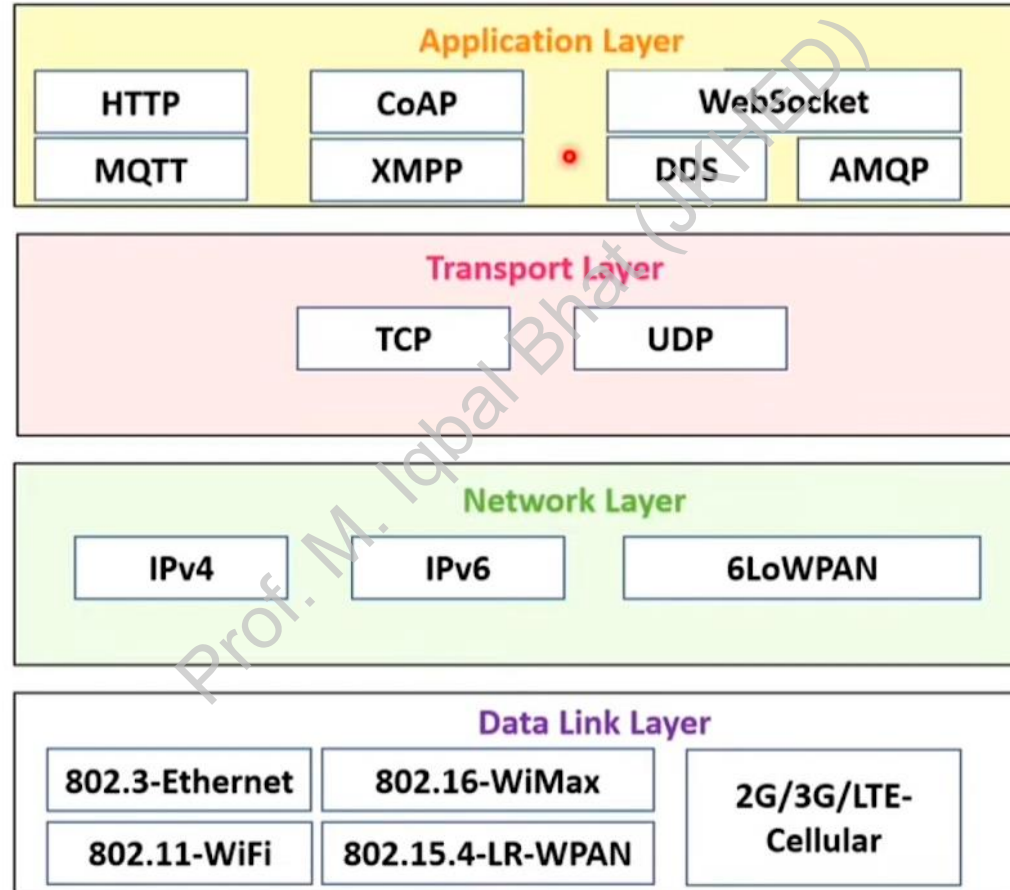
- Connectivity
 - WiFi
 - Blue Tooth
 - ZigBee
 - LoRa



Building blocks of IoT



Block diagram of Protocols



Link Layer

- Protocols determine how data is physically sent over the network's physical layer or medium.
- Local network connect to which host is attached.
- Hosts on the same link exchange data packets over the link layer using link layer protocols.
- Link layer determines how packets are coded and signaled by the h/w device over the medium to which the host is attached.

Link Layer

- **802.3-Ethernet:** IEEE802.3 is collection of wired Ethernet standards for the link layer. Eg: 802.3 uses co-axial cable; 802.3i uses copper twisted pair connection; 802.3j uses fiber optic connection; 802.3ae uses Ethernet overfiber.
- **802.11-WiFi:** IEEE802.11 is a collection of wireless LAN(WLAN) communication standards including extensive description of link layer. Eg: 802.11a operates in 5GHz band, 802.11b and 802.11g operates in 2.4GHz band, 802.11n operates in 2.4/5GHz band, 802.11ac operates in 5GHz band, 802.11ad operates in 60Ghzband.
- **802.16 - WiMax:** IEEE802.16 is a collection of wireless broadband standards including exclusive description of link layer. WiMax provide data rates from 1.5 Mb/s to 1Gb/s.
- **802.15.4-LR-WPAN:** IEEE802.15.4 is a collection of standards for low rate wireless personal area network(LR-WPAN). Basis for high level communication protocols such as ZigBee. Provides data rate from 40kb/s to250kb/s.
- **2G/3G/4G/5G-Mobile Communication:** Data rates from 9.6kb/s(2G) to up to100Mb/s(4G).

Network/Internet Layer

- Responsible for sending IP datagrams from source n/w to destination n/w.
- Performs the host addressing and packet routing.
- Datagrams contains source and destination address.

Network/Internet Layer

- **IPv4:** Internet Protocol version 4 is used to identify the devices on a n/w using a hierarchical addressing scheme. 32 bit address. Allows total of 2^{32} addresses.
- **IPv6:** Internet Protocol version 6 uses 128 bit address scheme and allows 2^{128} addresses.
- **6LOWPAN:** (IPv6 over Lowpower Wireless Personal Area Network) operates in 2.4 GHz frequency range and data transfer 250 kb/s.

Transport Layer

- Provides end-to-end message transfer capability independent of the underlying n/w.
- Set up on connection with ACK as in TCP and without ACK as in UDP.
- Provides functions such as error control, segmentation, flow control and congestion control.

Transport Layer

- **TCP:** Transmission Control Protocol used by web browsers(along with HTTP and HTTPS), email(along with SMTP, FTP). Connection oriented and stateless protocol. IP Protocol deals with sending packets, TCP ensures reliable transmission of protocols in order. Avoids n/w congestion and congestioncollapse.
- **UDP:** User Datagram Protocol is connectionless protocol. Useful in time sensitive applications, very small data units to exchange. Transaction oriented and stateless protocol. Does not provide guaranteeddelivery.

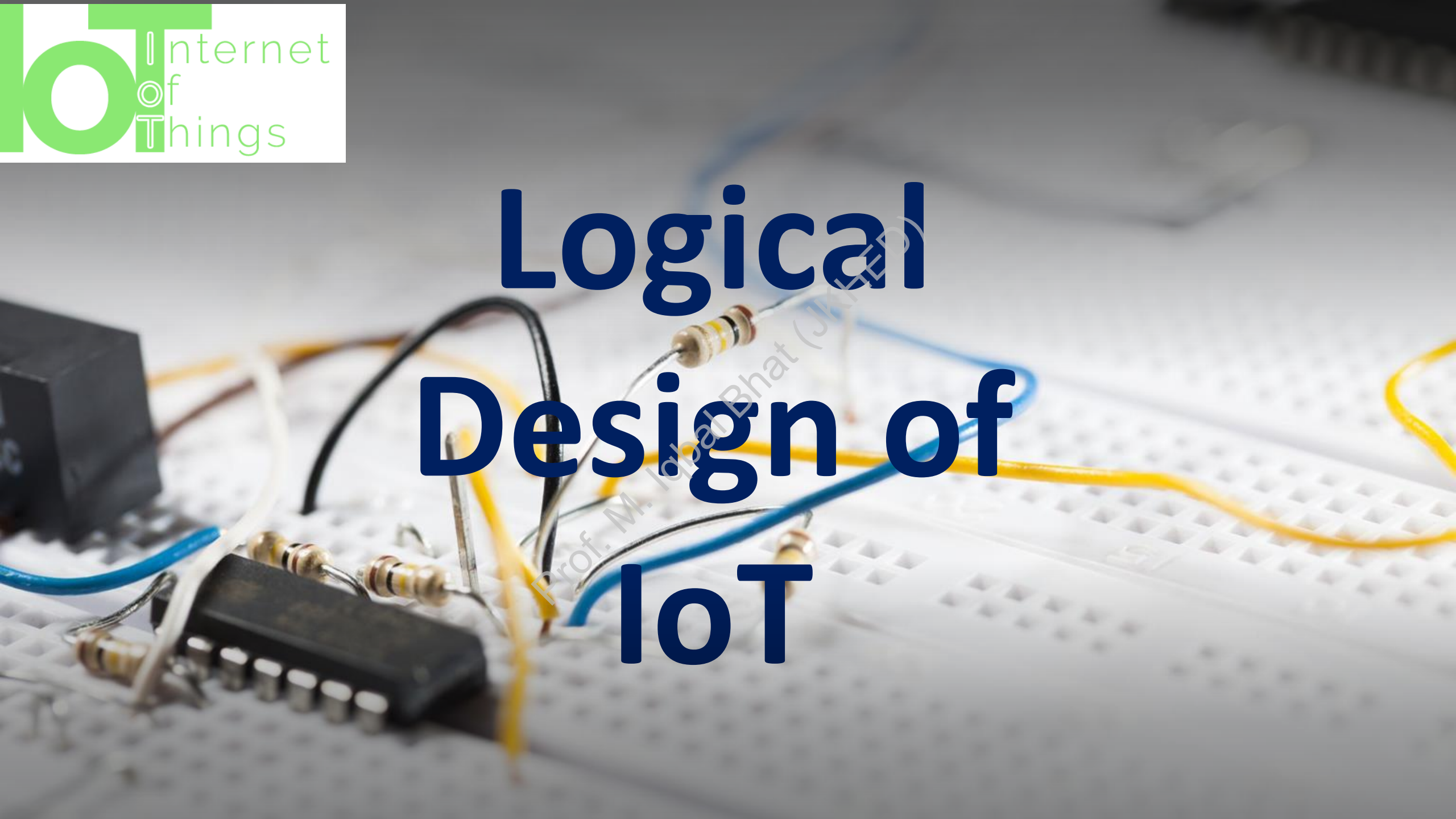
Application Layer

- Defines how the applications interface with lower layer protocols to send data over the n/w.
- Enables process-to-process communication using ports.

Application Layer

- **HTTP:** Hyper Text Transfer Protocol that forms foundation of WWW. Follow request- response model Stateless protocol.
- **CoAP:** Constrained Application Protocol for machine-to-machine(M2M) applications with constrained devices, constrained environment and constrained n/w. Uses client- server architecture.
- **WebSocket:** allows full duplex communication over a single socket connection.
- **MQTT:** Message Queue Telemetry Transport is light weight messaging protocol based on publish-subscribe model. Uses client server architecture. Well suited for constrained environment.
- **XMPP:** Extensible Message and Presence Protocol for real time communication and streaming XML data between network entities. Support client-server and server-server communication.
- **DDS:** Data Distribution Service is data centric middleware standards for device-to-device or machine-to-machine communication. Uses publish-subscribe model.
- **AMQP:** Advanced Message Queuing Protocol is open application layer protocol for business messaging. Supports both point-to-point and publish-subscribe model.

Logical Design of IoT

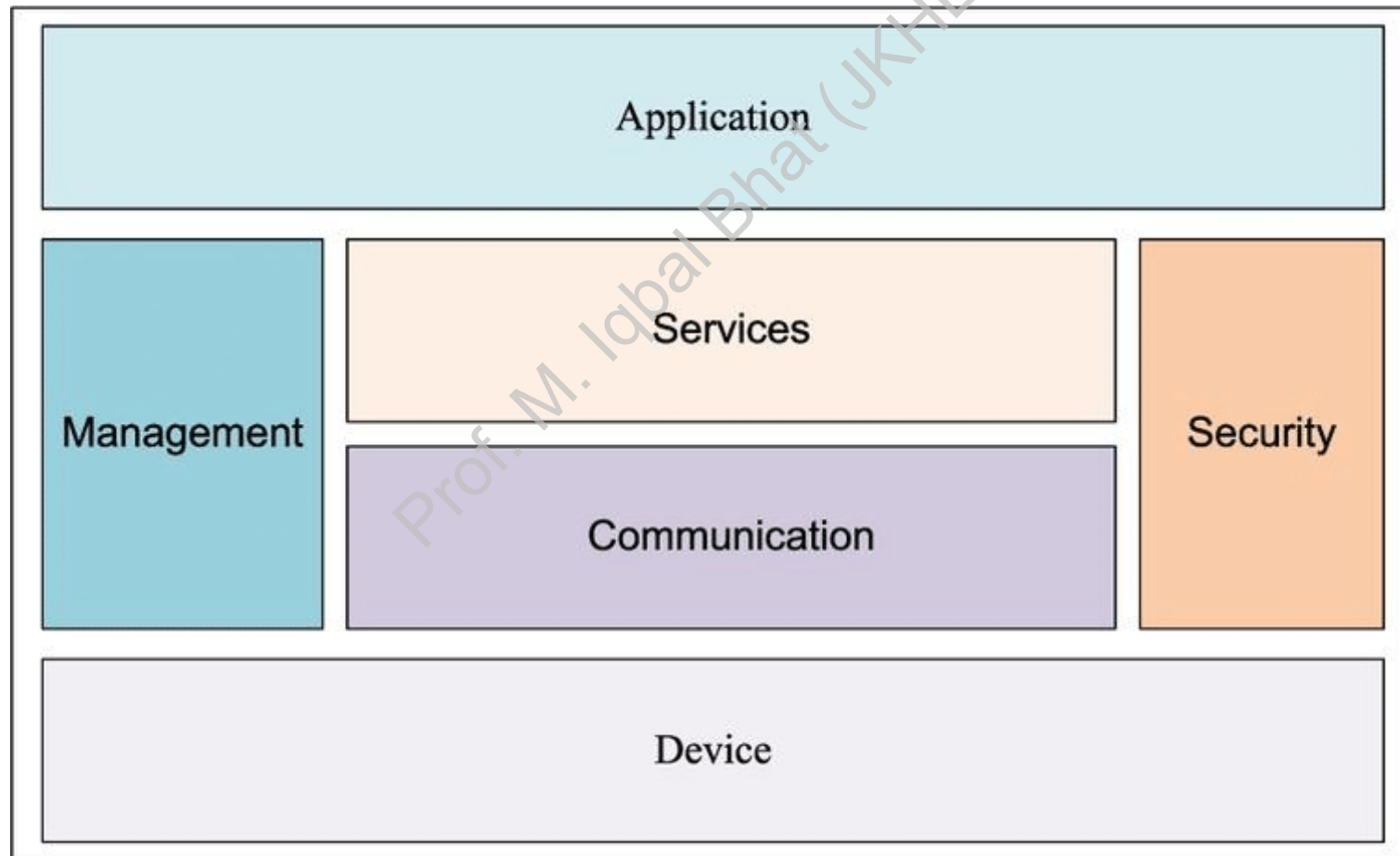
The background of the slide is a close-up photograph of a breadboard circuit. A black integrated circuit (IC) is plugged into the breadboard, with several resistors and jumper wires connected to it. The wires are color-coded, with blue and yellow being prominent. The breadboard itself is white with a grid of holes. The text 'Logical Design of IoT' is overlaid in the center in a large, bold, blue font. A faint watermark 'Prof. M. Rajesh Bhat (JNTU)' is visible diagonally across the center of the image.

Logical Design of IoT

- Refers to an abstract represent of entities and processes without going into the low level specifics of implementation.
- It includes:
 1. IoT Functional Blocks
 2. IoT Communication Models
 3. IoT Communication APIs

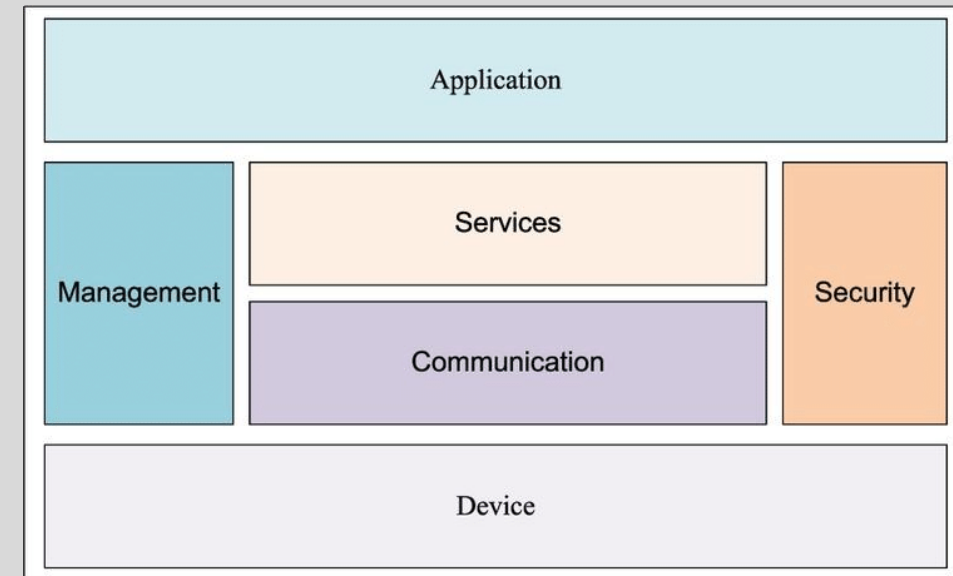
1. Functional Blocks of IoT

Provide the system the capabilities for identification, sensing, actuation, communication and management.



Functional Blocks of IoT

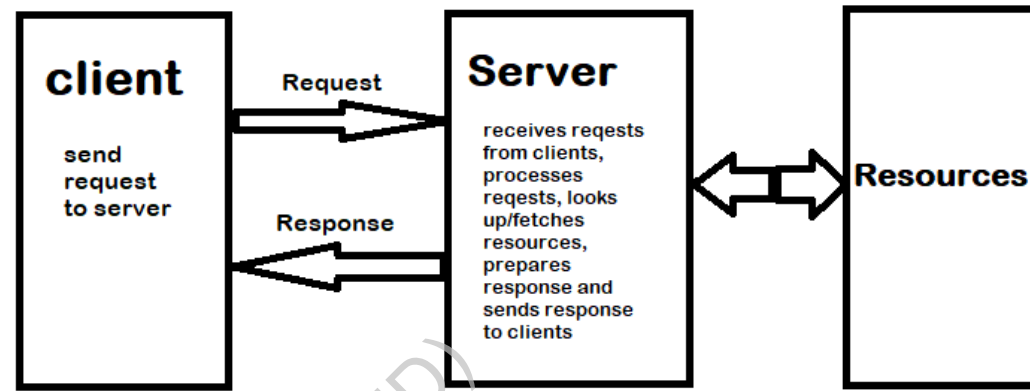
- **Device:** An IoT system comprises of devices that provide sensing, actuation, monitoring and control functions.
- **Communication:** handles the communication for IoT system.
- **Services:** for device monitoring, device control services, data publishing services and services for device discovery.
- **Management:** Provides various functions to govern the IoT system.
- **Security:** Secures IoT system and priority functions such as authentication, authorization, message and context integrity and data security.
- **Application:** IoT application provide an interface that the users can use to control and monitor various aspects of IoT system.



2. IoT Communication Model

- Communication models are key to how devices connect, interact, and share data in the Internet of Things (IoT).
- They determine efficiency, scalability, and overall system performance by defining the structure and flow of information inside an IoT network
 1. Request-Response Model
 2. Publish-Subscribe Model
 3. Push-Pull Model
 4. Exclusive Pair Model

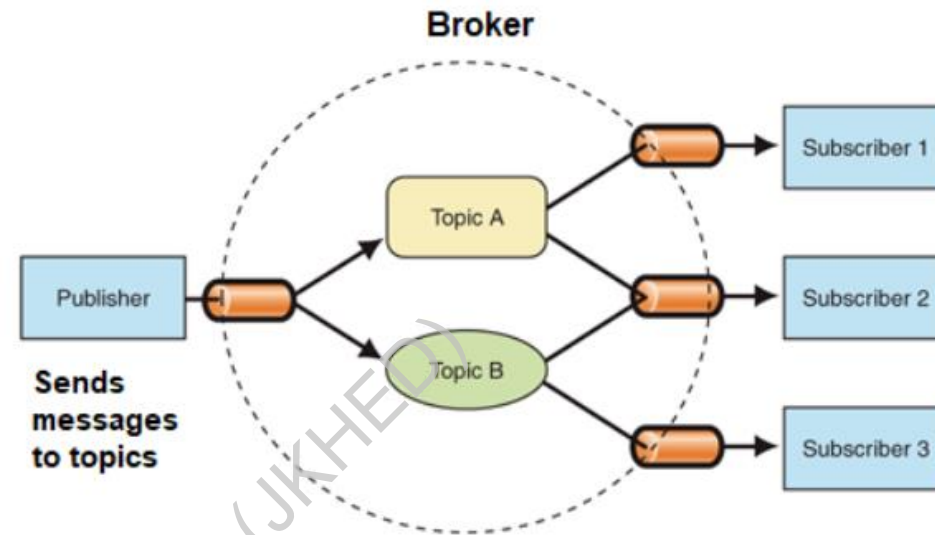
Request-Response Model



Request-Response Communication Model

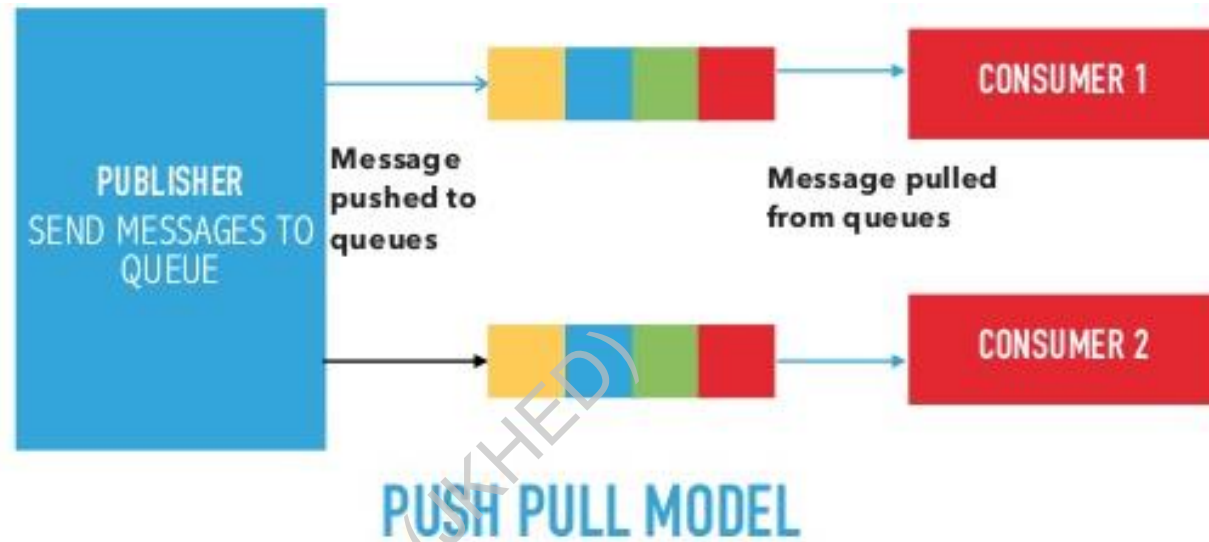
- Request-response model is communication model in which the client sends requests to the server and the server responds to the requests.
- When the server receives a request, it decides how to respond, fetches the data, retrieves resource representation, prepares the response, and then sends the response to the client.
- Request-response is a stateless communication model and each request-response pair is independent of others.
- HTTP works as a request-response protocol between a client and server. A web browser may be the client, and an application on a computer that hosts a web site may be the server.

Publish-Subscribe Model



- Publish-Subscribe is a communication model that involves publishers, brokers and consumers.
- Publishers are the source of data.
- Publishers send the data to the topics which are managed by the broker. Publishers are not aware of the consumers.
- Consumers subscribe to the topics which are managed by the broker.
- When the broker receive data for a topic from the publisher, it sends the data to all the subscribed consumers.

Push-Pull Model



- Push-Pull is a communication model in which the data producers push the data to queues and the consumers Pull the data from the Queues.
- Producers do not need to be aware of the consumers.
- Queues help in decoupling the messaging between the Producers and Consumers.
- Queues also act as a buffer which helps in situations when there is a mismatch between the rate at which the producers push data and the rate rate at which the consumer pull data.

Exclusive Pair Model



- Exclusive Pair is a bidirectional, fully duplex communication model that uses a persistent connection between the client and server.
- Connection is setup it remains open until the client sends a request to close the connection.
- Client and server can send messages to each other after connection setup.
- Exclusive pair is stateful communication model and the server is aware of all the open connections.

Choosing Right Communication Model:

Depends on

- **Network size and complexity:** Larger networks may benefit from publish-subscribe or push-pull models for scalability.
- **Data volume and frequency:** High-volume, real-time data might require push-pull or exclusive pair models.
- **Device capabilities and resource constraints:** Simple devices might be limited to client-server models.
- **Security and privacy considerations:** Some models offer better data isolation and control than others.

3. IoT Communication API's

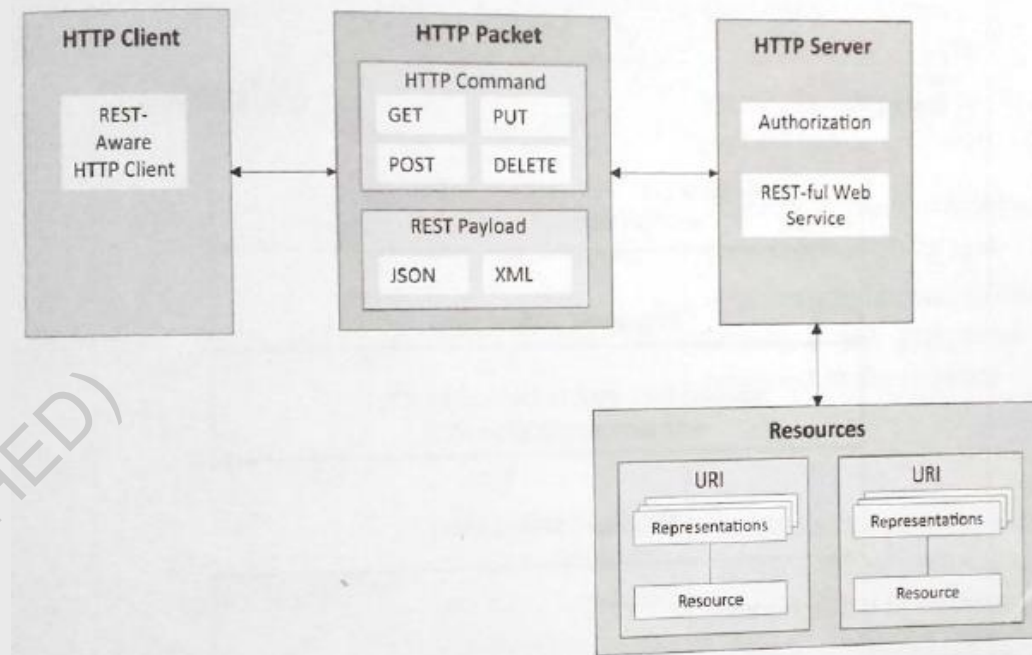
- IoT Communications APIs are a set of protocols and interfaces that allow IoT devices and platforms to communicate with one another.
- These APIs enable developers to create apps that interface with IoT devices over conventional web protocols including HTTP, MQTT, CoAP, and others.
- Types:
 1. REST based communication APIs(Request-Response Based Model)
 2. WebSocket based Communication APIs(Exclusive Pair Based Model)

3. IoT Communication API's

- **REST based communication APIs.**
- Representational State Transfer(REST) is a set of architectural principles by which we can design web services and web APIs that focus on a system's resources and have resource states are addressed and transferred..

REST based communication APIs

- **Client-Server:** The principle behind client-server constraint is the separation of concerns. Separation allows client and server to be independently developed and updated.
- **Stateless:** Each request from client to server must contain all the info. Necessary to understand the request, and cannot take advantage of any stored context on the server.
- **Cache-able:** Cache constraint requires that the data within a response to a request be implicitly or explicitly labeled as cache-able or non-cacheable. If a response is cache-able, then a client cache is given the right to reuse that response data for later, equivalent requests.
- **Layered System:** constraints the behavior of components such that each component cannot see beyond the immediate layer with which they are interacting.
- **User Interface:** constraint requires that the method of communication between a client and a server must be uniform.
- **Code on Demand:** Servers can provide executable code or scripts for clients to execute in their context. This constraint is the only one that is optional.



REST based communication APIs

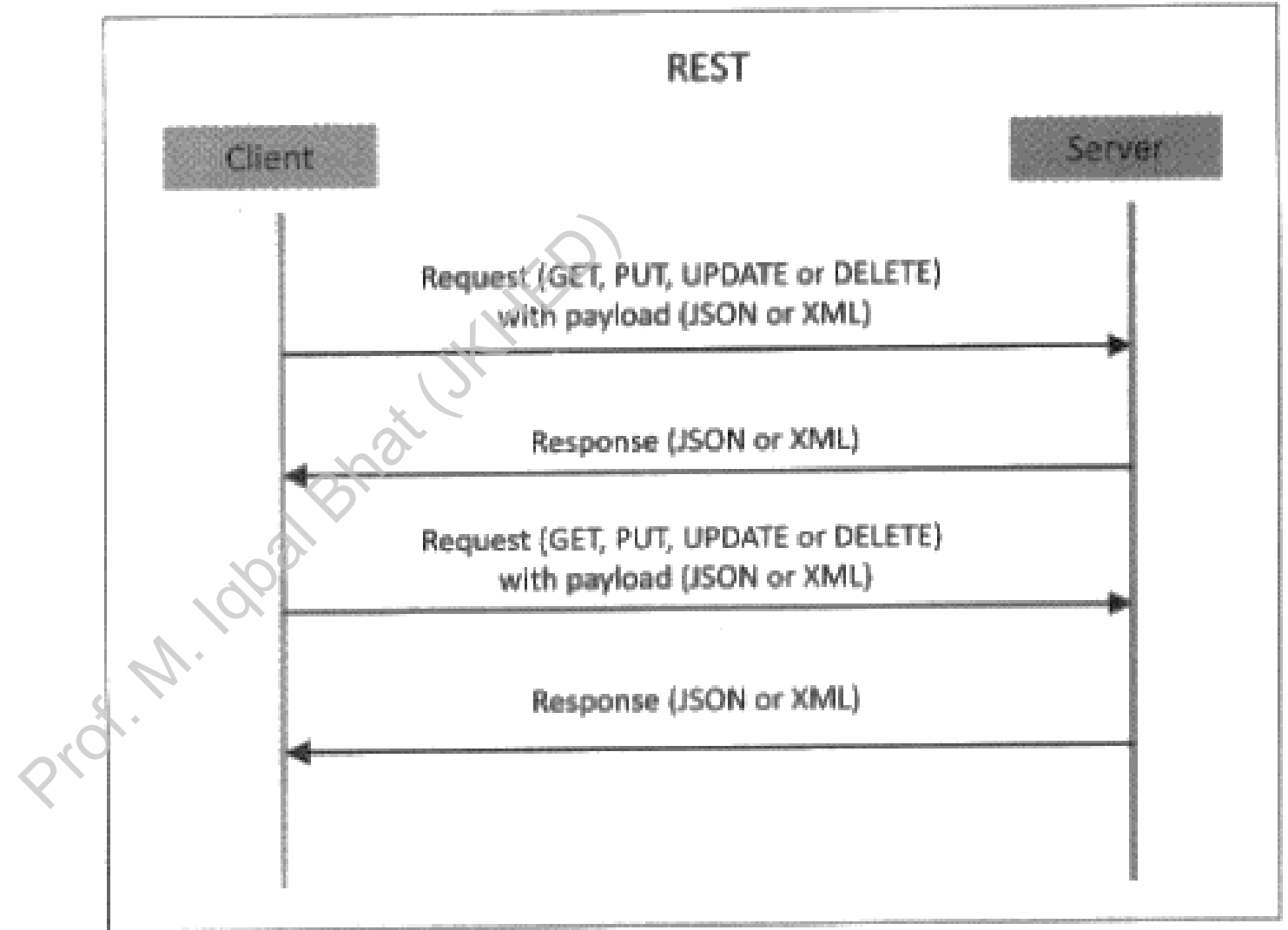
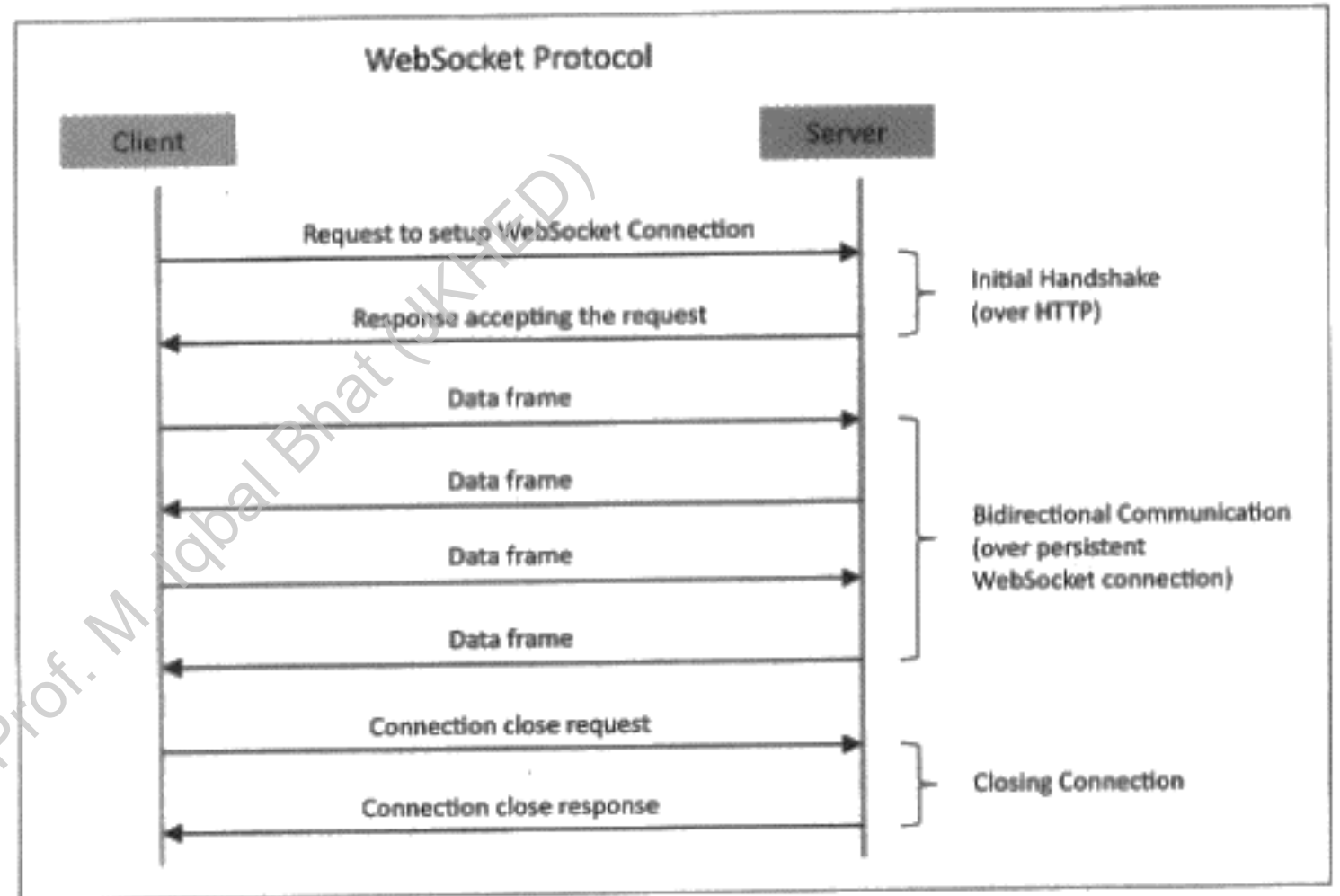


Figure 1.12: Request-response model used by REST

3. IoT Communication API's

- **WebSocket Based Communication APIs:**
- WebSocket APIs allow bi-directional, full duplex communication between clients and servers.
- WebSocket APIs follow the exclusive pair communication model.

WebSocket Based Communication APIs:



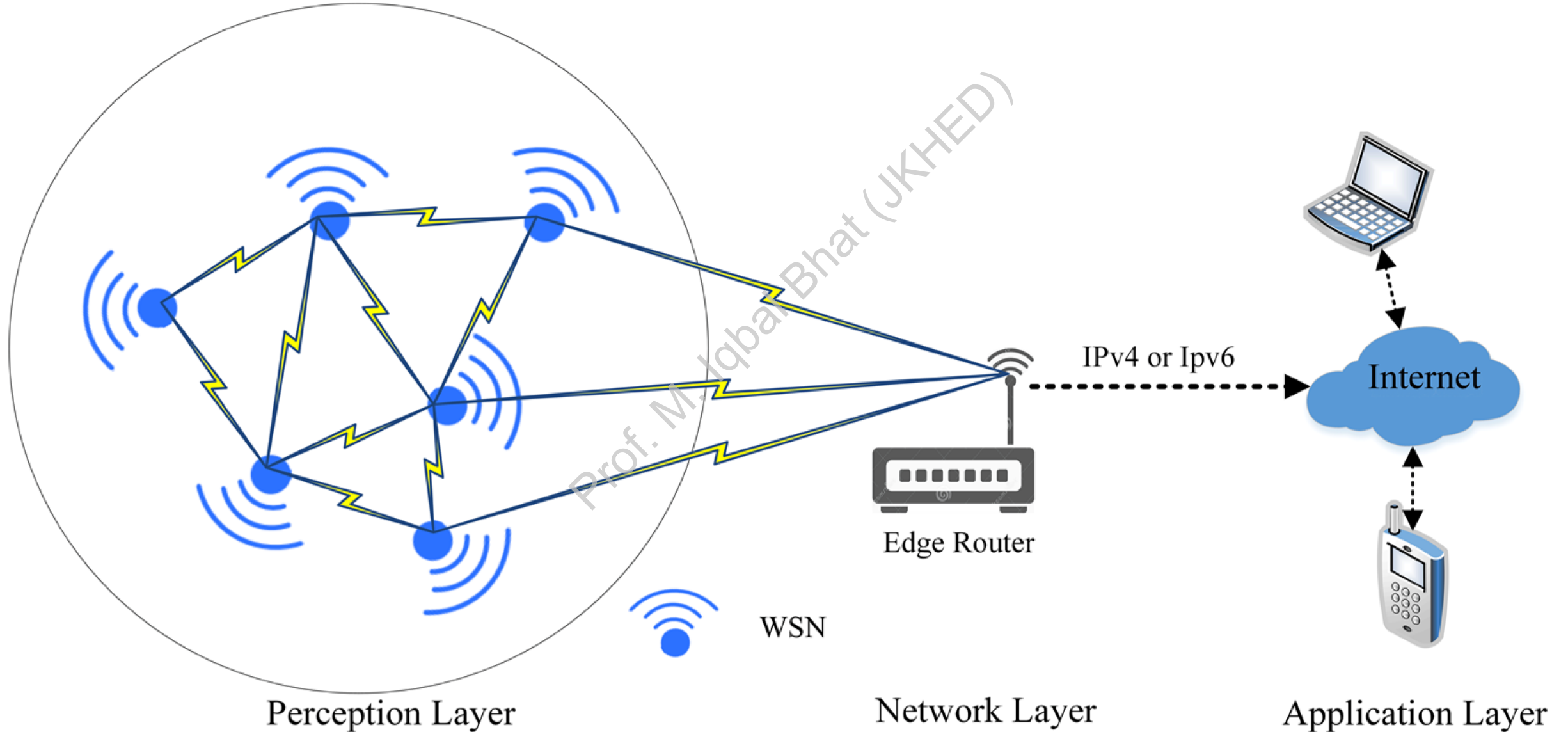
IoT Enabling Technologies

- IoT is enabled by several technologies including
 1. Wireless Sensor Networks,
 2. Cloud Computing,
 3. Big Data Analytics,
 4. Embedded Systems,
 5. Security Protocols and architectures,
 6. Communication Protocols,
 7. Web Services,
 8. Mobile internet and
 9. semantic search engines

Wireless Sensor Network(WSN)

- Comprises of distributed devices with sensors which are used to monitor the environmental and physical conditions. Zig Bee is one of the most popular wireless technologies used by WSNs.
- WSNs used in IoT systems are described as follows:
 - Weather Monitoring System: in which nodes collect temp, humidity and other data, which is aggregated and analyzed.
 - Indoor air quality monitoring systems: to collect data on the indoor air quality and concentration of various gases.
 - Soil Moisture Monitoring Systems: to monitor soil moisture at various locations.
 - Surveillance Systems: use WSNs for collecting surveillance data(motion data detection).
 - Smart Grids : use WSNs for monitoring grids at various points
 - Structural Health Monitoring Systems: Use WSNs to monitor the health of structures(building, bridges) by collecting vibrations from sensor nodes deployed at various points in the structure.

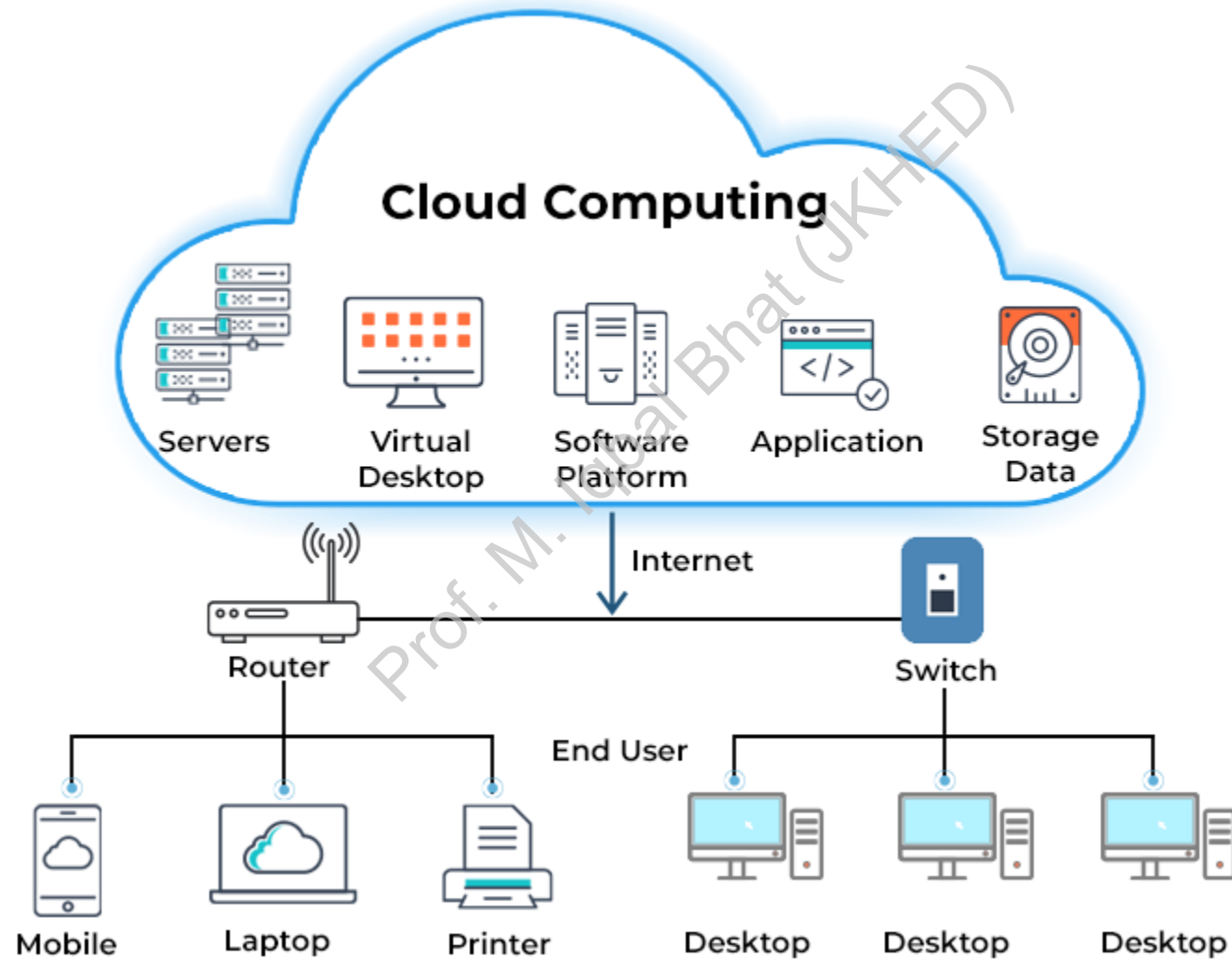
Wireless Sensor Network(WSN)



Cloud Computing

- **Infrastructure-as-a-service(IaaS)**:provides users the ability to provision computing and storage resources. These resources are provided to the users as a virtual machine instances and virtual storage.
- **Platform-as-a-Service(PaaS)**: provides users the ability to develop and deploy application in cloud using the development tools, APIs, software libraries and services provided by the cloud service provider.
- **Software-as-a-Service(SaaS)**: provides the user a complete software application or the user interface to the application itself..

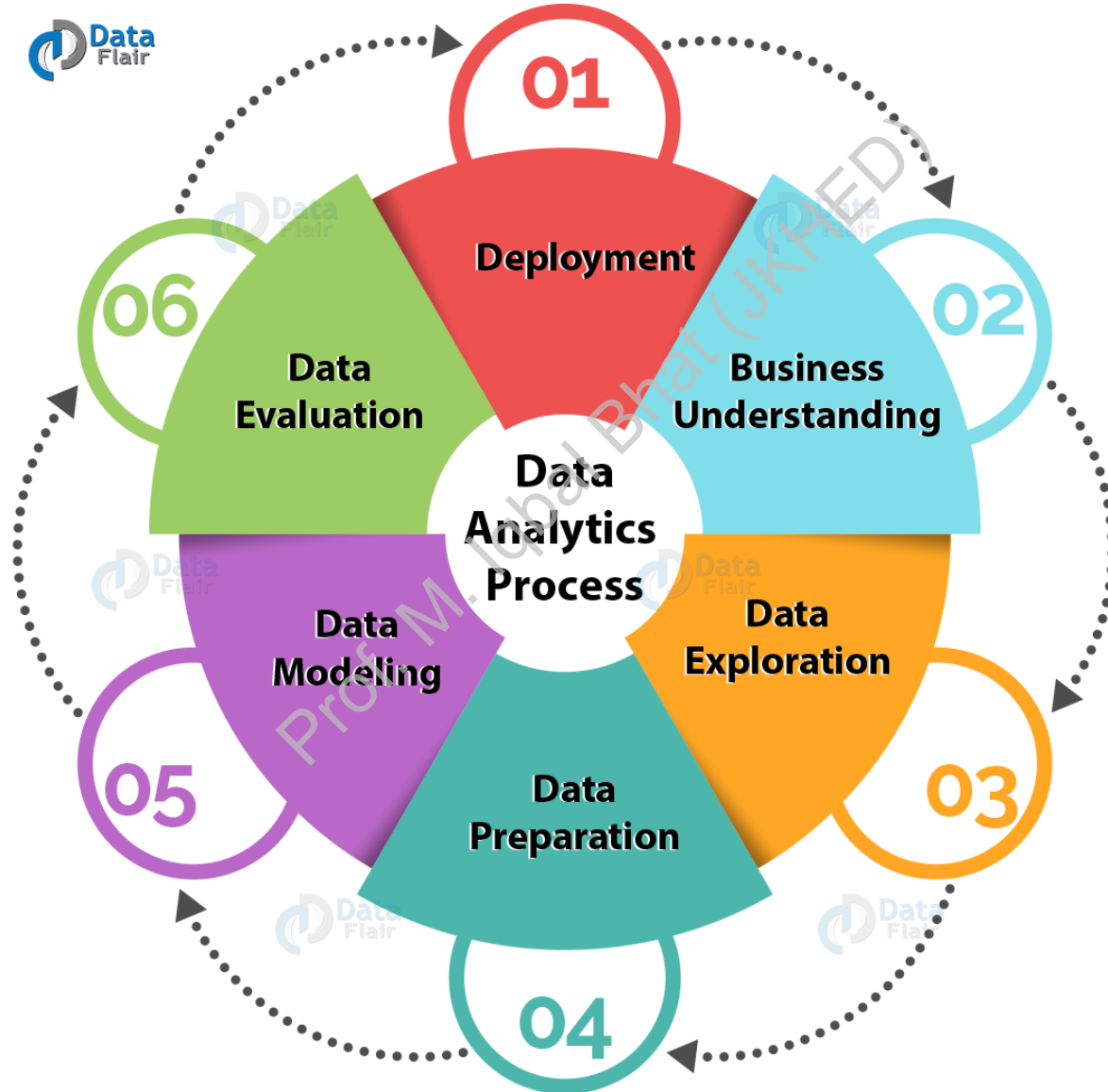
Cloud Computing



Big Data Analytics

- Sensor data generated by IoT systems.
 - Machine sensor data collected from sensors established in industrial and energy systems.
 - Health and fitness data generated IoT devices.
 - Data generated by IoT systems for location and tracking vehicles.
 - Data generated by retail inventory monitoring systems.

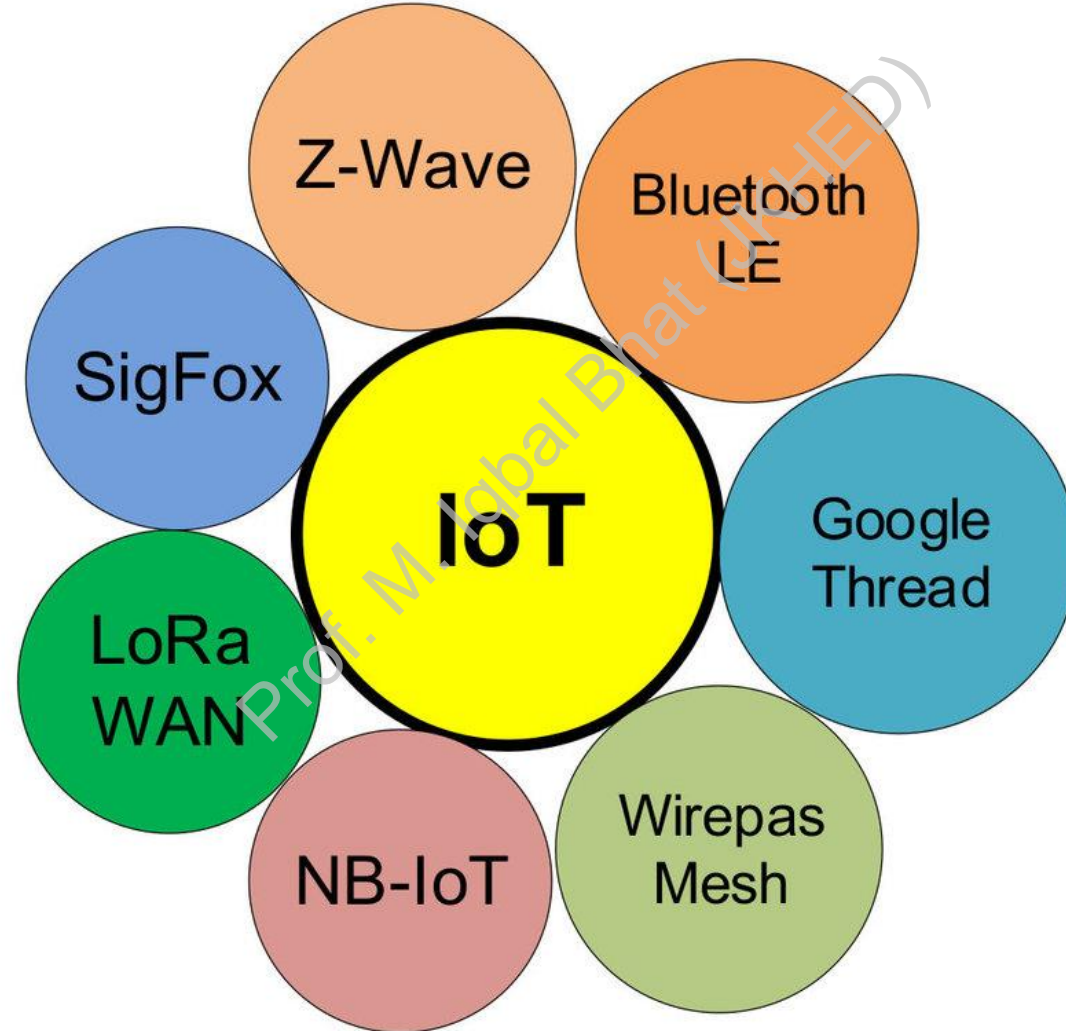
Big Data Analytics



Communication Protocols

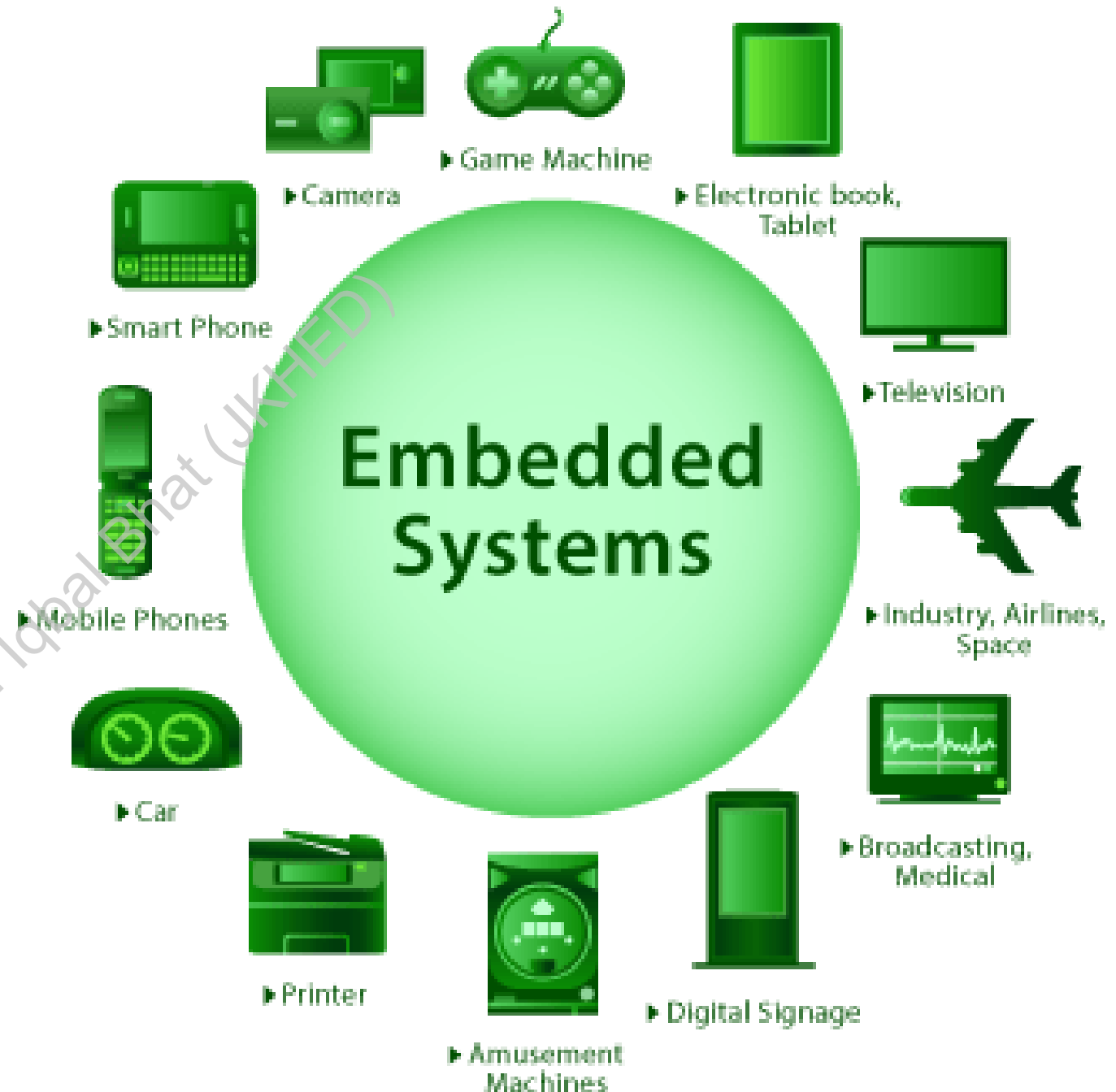
- form the back-bone of IoT systems and enable network connectivity and coupling to applications..
 - Allow devices to exchange data over network.
 - Define the exchange formats, data encoding addressing schemes for device and routing of packets from source to destination.
 - It includes sequence control, flow control and retransmission of lost packets

Communication Protocols



Embedded Systems

- It is a computer system that has computer hardware and software embedded to perform specific tasks. Embedded Systems range from low cost miniaturized devices such as digital watches to devices such as digital cameras, POS terminals, vending machines, appliances etc.





Thank You!

Prof. M. Iqbal Bhat (JKHED)